DB Schema / ERD v0.5 (Supabase)

Entities, attributes, relationships and primary/foreign key constraints (Supabase / PostgreSQL)

1. Entities and Attributes

1.1 auth.users (managed by Supabase)

Created and managed by Supabase Auth. It holds the user's UUID (`id`), email, encrypted password and `last_sign_in_at`. It is used for registration and login and is not created manually.

1.2 Profile (public.profiles)

Stores additional user information and the user's role.

Attribute	Data Type	Constraint	Description
id	uuid	PK, FK → auth.users(id) ON DELETE CASCADE	Same ID as the Supabase Auth user
full_name	varchar(100)	NOT NULL	Full name of the user
role	user_role (ENUM: STUDENT, ADMIN)	NOT NULL, DEFAULT 'STUDENT'	Determines access level
created_at	timestamptz	NOT NULL, DEFAULT now()	Profile creation time

1.3 Paper (public.papers)

Stores research papers added to the corpus by Admins.

Attribute	Data Type	Constraint	Description
id	uuid	PK, DEFAULT gen_random_uuid()	Unique identifier of the paper
external_id	varchar(255)	NOT NULL, UNIQUE	Paper ID or DOI from the corpus data source (prevents duplicates)
title	varchar(500)	NOT NULL	Title of the paper
authors	text	NULL	Author names
venue	varchar(255)	NULL	Journal or conference name
publication_year	int	NULL	Year of publication
source_url	varchar(500)	NULL	URL of the paper

Attribute	Data Type	Constraint	Description
pdf_path	varchar(500)	NULL	File path inside the Storage bucket `papers`
added_by	uuid	FK → profiles(id) ON DELETE RESTRICT, NOT NULL	Admin who added the paper
added_at	timestamptz	NOT NULL, DEFAULT now()	Time the paper was added

1.4 Rhetorical Move (public.rhetorical_moves)

Lookup table of rhetorical move categories used for tagging (for example Establish Territory, Identify Gap, Present Contribution).

Attribute	Data Type	Constraint	Description
id	int (identity)	PK	Unique identifier of the move
move_name	varchar(100)	NOT NULL, UNIQUE	Name of the rhetorical move
description	text	NULL	Explanation of the move
section_type	varchar(50)	NULL	Section it typically belongs to (e.g., Introduction)

1.5 Pattern (public.patterns)

Text segments extracted from papers by GROBID and tagged with a rhetorical move. They are used as examples when generating critiques.

Attribute	Data Type	Constraint	Description
id	uuid	PK, DEFAULT gen_random_uuid()	Unique identifier of the pattern
paper_id	uuid	FK → papers(id) ON DELETE CASCADE, NOT NULL	Paper the pattern was extracted from
move_id	int	FK → rhetorical_moves(id) ON DELETE RESTRICT, NOT NULL	Rhetorical move assigned to the text
text_excerpt	text	NOT NULL	The extracted sentence or passage
section_name	varchar(100)	NULL	Section of the paper it came from

Attribute	Data Type	Constraint	Description
sequence_no	int	NULL	Position of the excerpt within the paper
created_at	timestamptz	NOT NULL, DEFAULT now()	Time of extraction

1.6 Draft (public.drafts)

Stores draft text submitted by Students.

Attribute	Data Type	Constraint	Description
id	uuid	PK, DEFAULT gen_random_uuid()	Unique identifier of the draft
user_id	uuid	FK → profiles(id) ON DELETE CASCADE, NOT NULL	Student who submitted the draft
content	text	NOT NULL	The submitted draft text
submitted_at	timestamptz	NOT NULL, DEFAULT now()	Submission time

1.7 Critique (public.critiques)

Stores the critique generated by the LLM for a draft. Together these rows form the Student's critique history.

Attribute	Data Type	Constraint	Description
id	uuid	PK, DEFAULT gen_random_uuid()	Unique identifier of the critique
draft_id	uuid	FK → drafts(id) ON DELETE CASCADE, NOT NULL	Draft the critique belongs to
critique_text	text	NOT NULL	The critique returned by the LLM
llm_model	varchar(100)	NULL	Name of the LLM used
created_at	timestamptz	NOT NULL, DEFAULT now()	Time the critique was generated

1.8 Critique Pattern (public.critique_patterns)

Junction table that resolves the many-to-many relationship between critiques and patterns. It records which example patterns were used in the prompt for a given critique.

Attribute	Data Type	Constraint	Description
critique_id	uuid	PK (composite), FK → critiques(id) ON DELETE CASCADE	Critique reference

Attribute	Data Type	Constraint	Description
pattern_id	uuid	PK (composite), FK → patterns(id) ON DELETE CASCADE	Pattern reference

2. Relationships

#	Relationship	Cardinality	Description
1	auth.users – Profile	One-to-One	Each auth user has exactly one profile.
2	Profile (Admin) – Paper	One-to-Many	An Admin can add many papers; each paper is added by one Admin.
3	Paper – Pattern	One-to-Many	A paper contains many patterns; each pattern comes from one paper.
4	Rhetorical Move – Pattern	One-to-Many	A move can be assigned to many patterns; each pattern has one move.
5	Profile (Student) – Draft	One-to-Many	A Student can submit many drafts; each draft belongs to one Student.
6	Draft – Critique	One-to-Many	A draft can receive many critiques (e.g., after resubmission).
7	Critique – Pattern	Many-to-Many	A critique can use many patterns and a pattern can appear in many critiques, resolved through critique_patterns.

3. Primary and Foreign Key Constraints

Primary Keys

- profiles(id)
- papers(id)
- rhetorical_moves(id)
- patterns(id)
- drafts(id)
- critiques(id)
- critique_patterns(critique_id, pattern_id), a composite key

Foreign Keys

- profiles.id → auth.users.id (ON DELETE CASCADE)
- papers.added_by → profiles.id (ON DELETE RESTRICT)
- patterns.paper_id → papers.id (ON DELETE CASCADE)
- patterns.move_id → rhetorical_moves.id (ON DELETE RESTRICT)
- drafts.user_id → profiles.id (ON DELETE CASCADE)

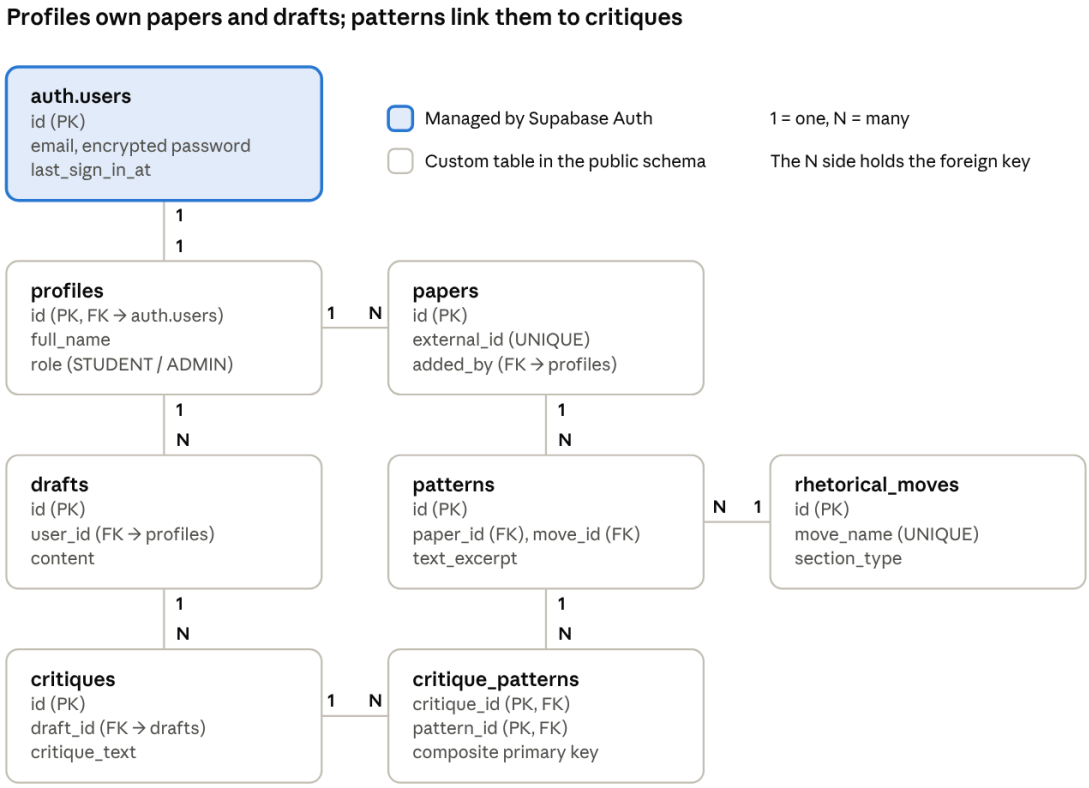
- critiques.draft_id → drafts.id (ON DELETE CASCADE)
- critique_patterns.critique_id → critiques.id (ON DELETE CASCADE)
- critique_patterns.pattern_id → patterns.id (ON DELETE CASCADE)

Other Integrity Constraints

- papers.external_id is UNIQUE. This supports the duplicate paper exception in the Add Paper sequence diagram (400 Bad Request).
- rhetorical_moves.move_name is UNIQUE.
- profiles.role is restricted to STUDENT or ADMIN. Only Admins may add papers.
- Email uniqueness and password hashing are handled by Supabase Auth.
- All attributes marked NOT NULL must contain a value.

4. ERD Diagram

The Entity Relationship Diagram below shows all tables, their key attributes and the relationships between them.



ERD · 7 custom tables plus auth.users, 8 relationships

Admin profiles add papers, papers yield patterns, and each critique records the patterns it used through the junction table critique_patterns.